

Exp No:2 Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Aim:

To upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Description:

Use sample data set sales-data.csv.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
file_path = 'sales-data-new.csv'
df = pd.read_csv(file_path)

# Display first five rows
print(df.head())

# Check for missing values
print(df.isnull().sum())

# Fill missing Sales values with mean
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())
```

```
# Drop rows with missing important values
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
```

```
# Summary statistics
print(df.describe())
```

```
# Group by Product and calculate total Sales and Quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
```

```
print(product_summary)
```

```
# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
```

```
# Convert Date into datetime format
df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
```

```
# Line plot of sales over time
```

```
sales_over_time = df.groupby('Date').agg({
    'Sales': 'sum'
}).reset_index()

plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.show()

# Pivot table to analyze sales by Region and Product
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc='sum',
    fill_value=0
)

print(pivot_table)

# Correlation matrix
correlation_matrix = df[['Sales', 'Quantity']].corr()
```

```
print(correlation_matrix)
```

```
# Heatmap of correlation matrix
```

```
plt.figure(figsize=(8, 6))
```

```
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
```

```
plt.title('Correlation Matrix')
```

```
plt.show()
```

Output:

	Date	Product	Sales	Quantity	Region
0	01-01-2023	Product A New	255.0	14	North
1	02-01-2023	Product B New	197.5	13	South
2	03-01-2023	Product A New	278.0	15	North
3	04-01-2023	Product C New	370.0	16	East
4	05-01-2023	Product B New	232.0	14	West

Date 0

Product 0

Sales 0

Quantity 0

Region 0

dtype: int64

	Sales	Quantity
count	16.000000	16.000000
mean	298.125000	15.375000
std	73.635929	1.746425
min	197.500000	13.000000
25%	240.625000	14.000000

50% 283.750000 15.500000

75% 372.875000 17.000000

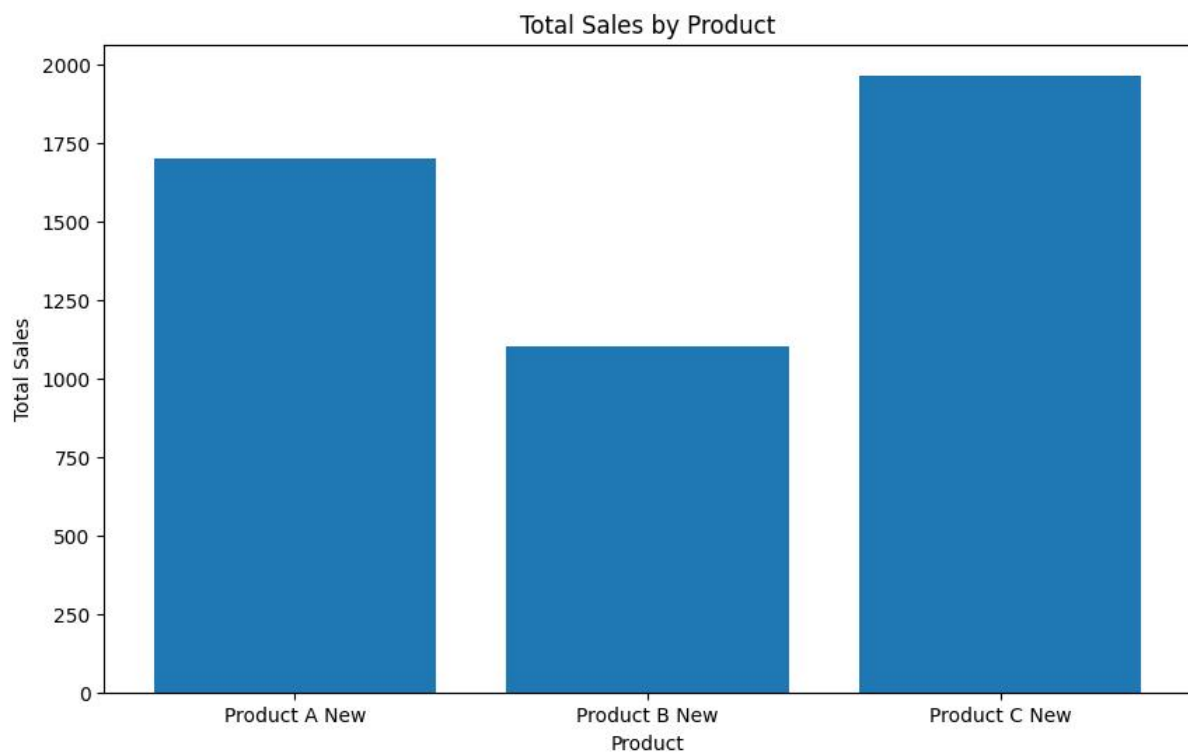
max 416.000000 18.000000

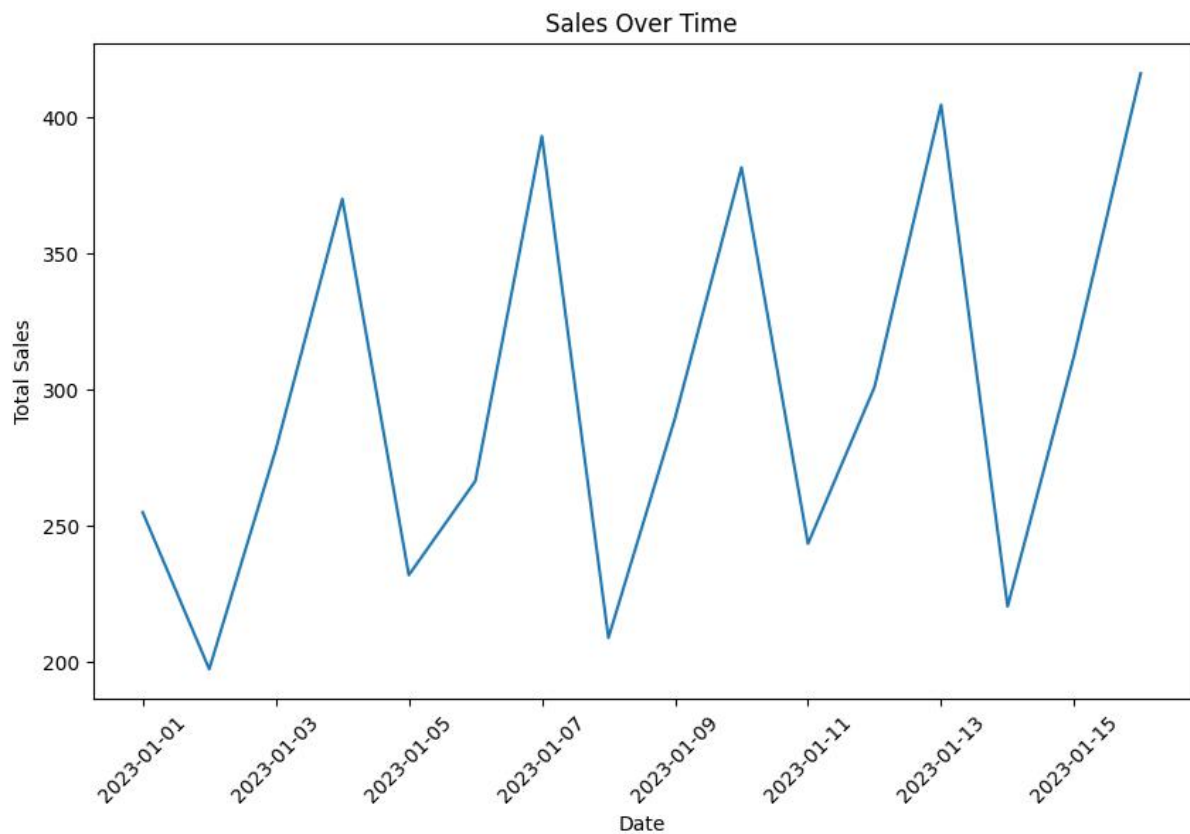
Product Sales Quantity

0 Product A New 1702.5 93

1 Product B New 1102.5 67

2 Product C New 1965.0 86





Product Product A New Product B New Product C New

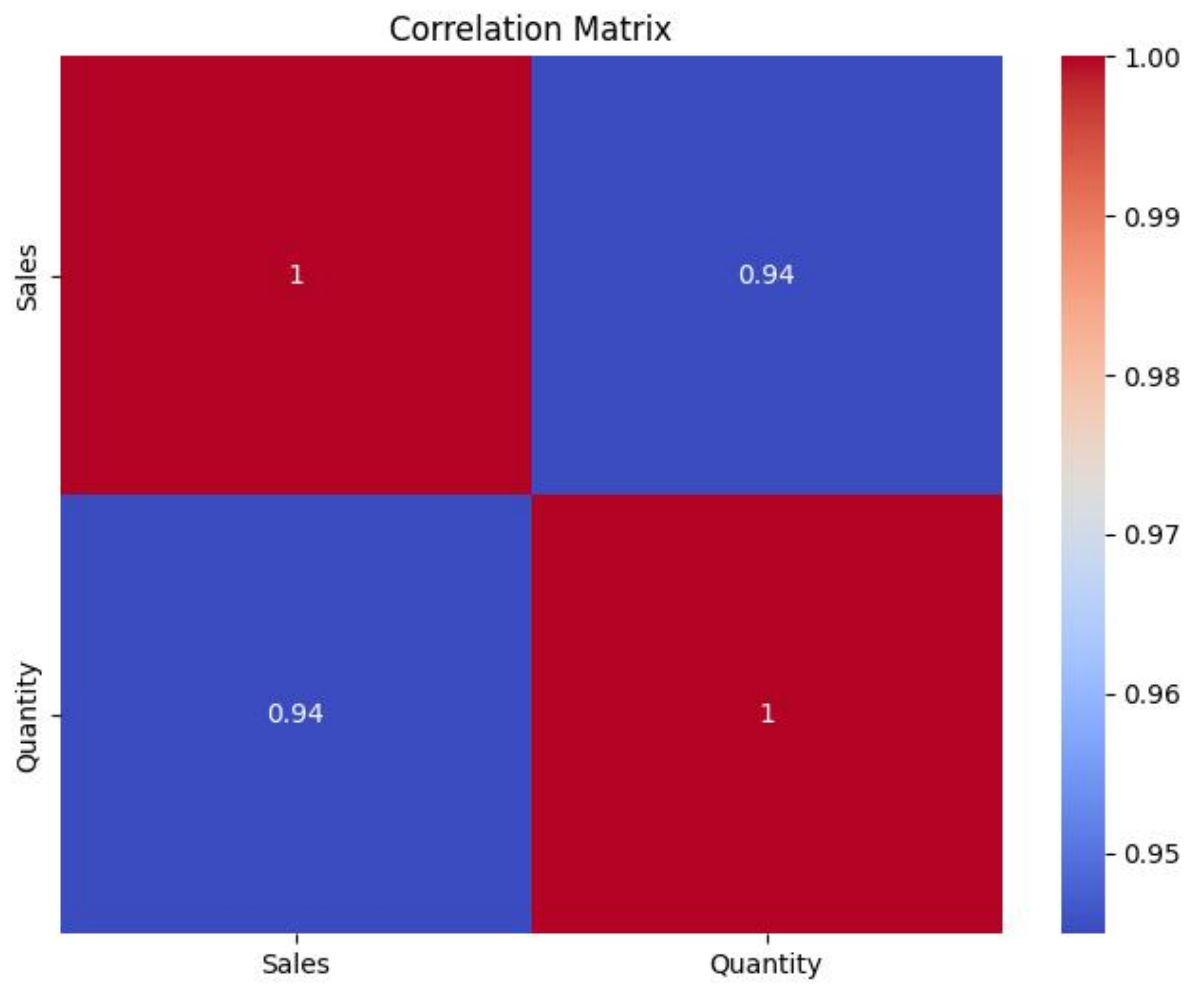
Region

East	0.0	0.0	1965.0
North	1702.5	0.0	0.0
South	0.0	627.0	0.0
West	0.0	475.5	0.0

Sales Quantity

Sales	1.000000	0.944922
-------	----------	----------

Quantity	0.944922	1.000000
----------	----------	----------



Result :